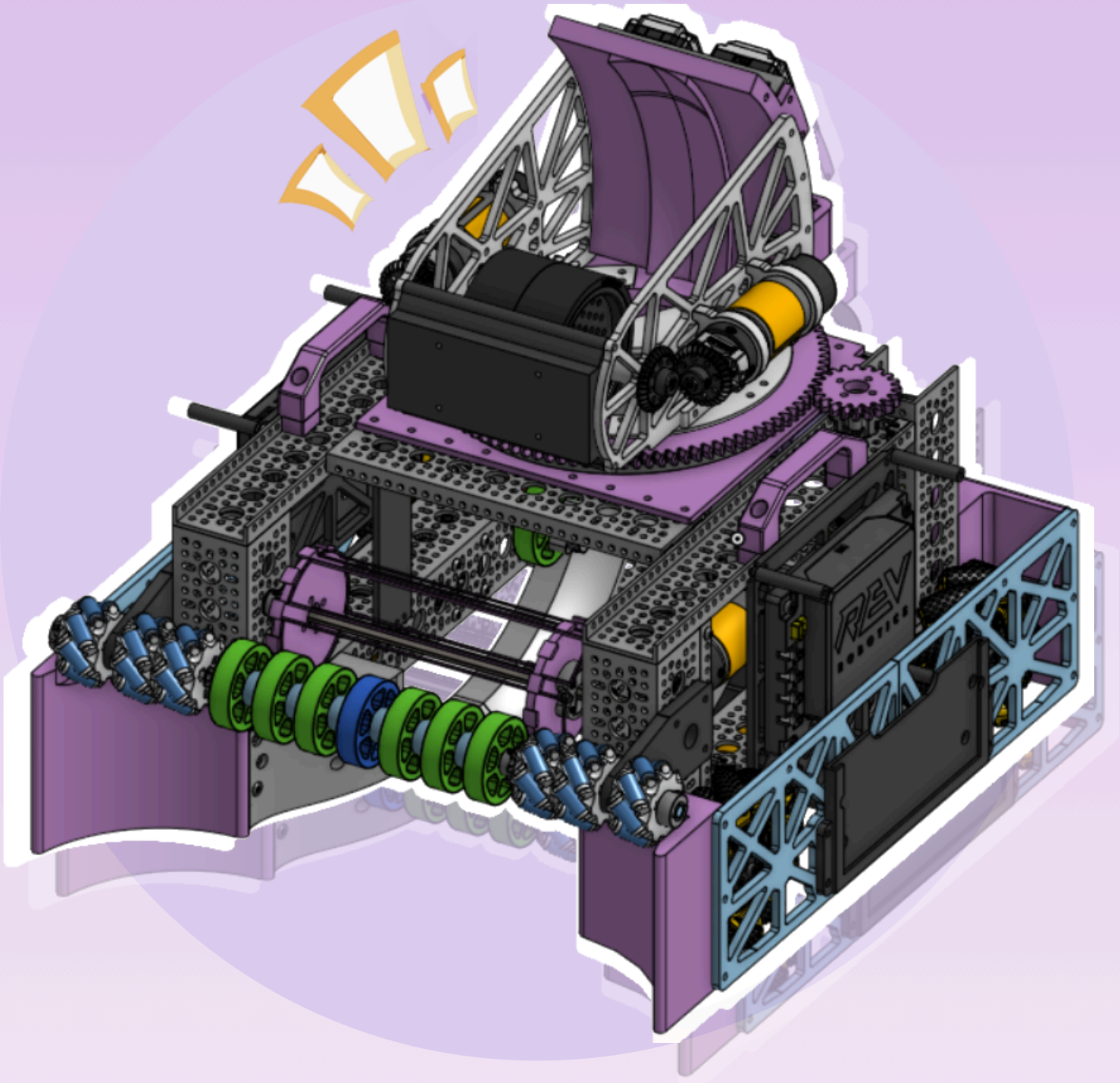


One Team
One Dream

FTC 51: Singularity

Engineering Portfolio 2025-26

INTRODUCTION Pg.1



Hello, we're **FTC 51 NerdHerd Singularity**. Our team consists of **5 sophomore/junior leaders** and **13 freshmen members** located in the **California Academy of Mathematics and Science** in **Carson, California**. United by our passion for **STEM** and **robotics**, we discovered our love for **innovation** through **teamwork** and **creativity**. Together, we are committed to sharing our enthusiasm for **FIRST**, **inspiring others**, and **celebrating every milestone** throughout our journey. With the support of our mentors and sponsors, we strive to embody **FIRST's core values** by being **inclusive teammates** and **respectful competitors** at every match. To keep our work neat, our team is organized into three sub-teams: **CAD, Programming, and Admin**. Leadership is shared between leads, who put emphasis on helping us **grow**.

VISION STATEMENT

Our vision is to build a thriving, **inclusive FIRST** community where new members evolve into **confident innovators,** mastering **technical skills**, **leadership**, and **collaboration**. Guided by **Gracious Professionalism** and **Coopertition,** we **empower** every member to reach their **full potential** through teamwork, project management, **inspiring a legacy** full of **STEM contributors.**

DEVELOPMENT PLAN

During our 2025–2026 season, our freshman members were introduced to **FTC** and the **FIRST organization** through a **structured onboarding process** designed to give them a clear understanding of both the program's mission and its technical expectations. Team leads organized a series of **hands-on workshops** covering **CAD, programming, and administrative operations,** allowing new members to gain practical experience with the tools and workflows used throughout the season. This foundation allowed members to explore multiple disciplines before **selecting a sub-team** that best matched their **interests and strengths,** where they continued developing skills in their chosen focus area.

Throughout the process, team leads **gradually increased task complexity,** to ensure each member felt supported while building confidence before transitioning to **independent work.** This approach helped students develop not only **technical proficiency** but also **problem-solving skills and accountability** within their sub-teams.

By consistently emphasizing **Gracious Professionalism, Coopertition, and the FIRST Core Values,** we fostered a culture of **respect, teamwork, communication, and integrity.** As a result, every member developed a **well-rounded understanding** of how technical skills and collaborative practices come together to drive success within the FIRST program and beyond. Our team organized the season using a prehistoric-era theme inspired by DECODE to help keep the team organized, aligned and motivated.

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MISSION STATEMENT

As a team with little to no experience in the FIRST robotics program, our mission is to build strong **relationships** and help each member develop their skills to their **fullest potential**. Guided by the FIRST core values, **Gracious Professionalism** and **Cooperation,** we promote **inclusion, teamwork**, and **mutual respect** regardless of their experience or background. Through thoughtful planning, clear goals, and high-quality standards, we **strengthen communication** and **project-management skills** across all our team members.

SPONSORS

Parker
Raising Cane's Chicken Fingers
HAAS
DOING GOOD WORKS
CADflow.ai
Raytheon
HONDA
BOEING

MEET THE DINOS

No Dino leads alone, the Herd makes the journey possible.



Charlee - 9th grade
Roles: CAD, Pit Crew, Judging Team
Favorite Dino: Dilophosaurus



Miley - 9th grade
Roles: CAD, Pit Crew
Favorite Dino: Plesiosaur



Lucas - 9th grade
Roles: CAD, Pit Crew
Favorite Dino: Velociraptor



Liam - 9th grade
Roles: CAD, Human Player, Judging Team
Favorite Dino: Archaeopteryx



Noah - 9th grade
Roles: CAD, Match Scout
Favorite Dino: Velociraptor



Elise - 9th grade
Roles: Programming, Pit Crew, Judging Team
Favorite Dino: Stegosaurus



Rui - 9th grade
Roles: Programming, Pit Scout
Favorite Dino: Brachiosaurus



Maxwell - 9th grade
Roles: Programming, Match Scout
Favorite Dino: Pterodactyl



Yeonah - 9th grade
Roles: Programming, Match Scout
Favorite Dino: Velociraptor



Victoria - 9th grade
Roles: Admin, Pit Crew
Favorite Dino: Plesiosaur



Manahil - 9th grade
Roles: Admin, Pit Crew, Judging Team
Favorite Dino: Archaeopteryx



Nadiya - 9th grade
Roles: Admin, Pit Scout
Favorite Dino: Plesiosaur



Honor - 9th grade
Roles: Admin, Match Scout
Favorite Dino: Pterodactyl



Marianne - 10th grade
Roles: CAD Lead, Pit Crew, Battery and Subsystem Manager
Favorite Dino: Stegosaurus



Vuth - 10th grade
Roles: CAD Lead, Match Scout Lead
Favorite Dino: Stegosaurus



Shassh - 11th grade
Roles: Programming Lead, Driver, Judging Team
Favorite Dino: Velociraptor



Brendan - 10th grade
Roles: Programming Lead, Match Scout
Favorite Dino: Pterodactyl



Dylan - 10th grade
Roles: Admin Lead, Pit Scout
Favorite Dino: Velociraptor



Jiwon Jung - 11th grade
Roles: Captain, Drive Coach
Favorite Dino: Tyrannosaurus Rex



Nathan Dunn - 11th grade
Roles: Captain
Favorite Dino: Velociraptor



Barney
Roles: MASCOT
Favorite Dino: Himself!



Mr. Lim
Roles: Mentor
Favorite Dino: Tyrannosaurus Rex



Mr. Hildebrandt
Roles: Mentor
Favorite Dino: Ankylosaurus



Mr. Umamaheswaran
Roles: Mentor
Favorite Dino: Stegosaurus

LESSONS LEARNED Pg.3

MEET REFLECTIONS

	What went well	What could be improved	Next steps
Meet 0	Shooting was quick and our robot was light and fast. We achieved third place in the competition.	Autos malfunctioned. Our intake would get stuck. Shooter aim was inconsistent.	Shooter design changed. We switched to Pedropathing for autos. More drive practice.
Meet 1	Human player was reliable. Shooter had adjustable distance. Auto was consistent.	Adjustable distances were inconsistent. Battery was poorly mounted.	We decided to split up our team into two in order to work on the new robot and iterate the old one.
Meet 2	Drive team was more experienced, had a 28 point auto and the shooter was more consistent.	We needed to communicate with our alliance partners more and our intake was inconsistent as well as adding more defense to our robot.	We bundled our wires together and CADded our design to make it more flushed and considering adding funneling at the point of the shooter.
Meet 3	We were able to fully rapid fire for scoring and had a successful red alliance auto.	Encountered mechanical issues, such as wires simply unplugging. Lacking surface to open the gate.	We need to complete turret auto tracking and make the front wheel guards higher with the side plates.

Meet	Match Points	Base Points	Auto Points	WLD
0	22.80	7.00	14.00	3-2-0
1	28.60	6.00	7.80	1-3-1
2	89.80	8.00	28.20	4-1-0
3	74.80	8.00	21.40	3-1-1

Performance Analysis

- Season Improvement: +294% from Meet 0 to Meet 2 (22.80 to 89.80 pts)
- Auto Points Growth: Doubled from 14.00 to 28.20 - our strongest category improvement
- Peak Performance: Meet 2 - highest score (89.80) and best record (4-1-0, 80% win rate)
- Consistency: Base points stable ranging from 6 to 8, showing reliable TeleOp execution
- Season Average: 53.95 match points across all meets
- Key Insight: Meet 1 dip (28.60) reflects strategic adjustments that paid off in later competitions

CAD Lead Reflections



MARIANNE

"I really liked this season so far. I've been able to learn and build not only robots, but friends! Although we had our **challenges**, overcoming them of course comes with time. I LOVE THIS FIRST, AND THIS **TEAM!**"



VUTH

"This season taught me not only **technical skills** but it further taught me how to apply these skills in a short time span. Furthermore this season taught me how to **collaborate**, and **communicate** in a large team environment. I loved the experience **Team 51** gave me and I will be applying the lesson I learned from this season into my future. Let herd together to form **singularity!**

Prog Lead Reflections



SHASSH

"This season was an **invaluable experience**, as I was able not only hone my technical skills, but also understand how to **manage, coordinate, and encourage collaboration**. I'm super excited to transfer these **skills** to new experiences throughout my life, and I have Team 51 to thank for such a **wonderful experience!**



BRENDAN

"This season was a wonderful year for me. I enjoyed teaching my members about Java and **building relationships!** In the future, I would like to refine my **programming skills** regarding localization. Overall, it was fun working with 51 again!"

Admin Lead Reflections



DYLAN

"This season is definitely one to look back on. I was able to teach my members about **outreach, portfolio, and team branding!** I was able to learn life skills like **leadership and communication** by teaching others. I loved being part of 51 Singularity and meeting new people! **One team, one dream!**"

SUSTAINABILITY PLAN

Season Goals

Goal 1: Build Strong Mentor and Professional Relationships

Objective: Build strong mentor and professional relationships

S - Connect with at least 3 STEM Professionals
M - Track number of meetings held.
A - Leverage existing networks
R - Strong relationships provide technical guidance + career insights
T - Complete before ILT (January)

Goal 2: Develop a Robust DFMA Design Process

Objective: Create/refine a repeatable design process using Design for Manufacturing and Assembly (DFMA) Principles

S - Document a step by step design process incorporating DFMA
M - Evaluate at least 3 designs using DFMA checklists
A - Build on existing knowledge through brainstorming and mentor guidance
R - Efficient design process improves robot quality and team productivity
T - Finalize process by September and apply throughout build season

Goal 3: Design a Competitive Robot

Objective: Design, Build, and Program a consistent and competitive robot

S - Create a robot that scores in the top 25% of scrimmages
M - Track iterations, test results, and competition ranking
A - Apply team skills, resources, and design process
R - Directly supports competition success
T - Optimize robot build throughout the season

Financial Sustainability

Annual Budget: $10,000

Income Sources:

- Parker - $800
- Raising Canes - $1500
- Honda - $1400
- Boeing - $1600
- Raytheon - $1300
- Doing Good Works - $1300
- HAAS- $1200
- Cadflow.ai - $900

Bill of Materials: Each time we needed to purchase materials, we organized a bill of materials (BOM) of such materials and their costs. By doing this, we managed to keep track of both our materials and expenses, leaving funds behind for future purchases.

Risk Management

Identified Risks

1. Loss of major sponsors
2. Lack of meeting space
3. Equipment failure/damage
4. High member turnover

Past Challenges

1. Insufficient sense of urgency
2. Poor deadline setting/execution
3. Low meeting productivity

→

Mitigation Strategies

1. List of potential backup sponsors
2. Utilize public spaces or members homes
3. (in BOM) Order extra parts
4. Established succession system (see page 1)

Past Challenges

1. Established clear timeline for season and worked on pace or ahead of said timeline
2. Utilized tools such as Gantt Chart and Kanbanchi; Stuck to plan no matter what
3. Ensured there was work for everyone; assigned members to leads

What is SMART Goal Setting?

SMART goal setting is a method used to ensure that goals are clear, realistic, and trackable. Each goal is evaluated using five criteria.

- **S** - **Specific:** Clearly defines what needs to be accomplished
- **M** - **Measurable:** Progress towards goals can be actively tracked
- **A** - **Achievable:** Goals are realistic, given team resources and constraints
- **R** - **Relevant:** Directly attribute towards team success
- **T** - **Time based:** Goal has a set/strict deadline

Using **SMART** goal setting helps our team to stay organized, measure progress, and ensure long term sustainability beyond just a single season.

In addition, by using measurable milestones, we are able to remain productive without risking burnout over time.

Progress Tracking

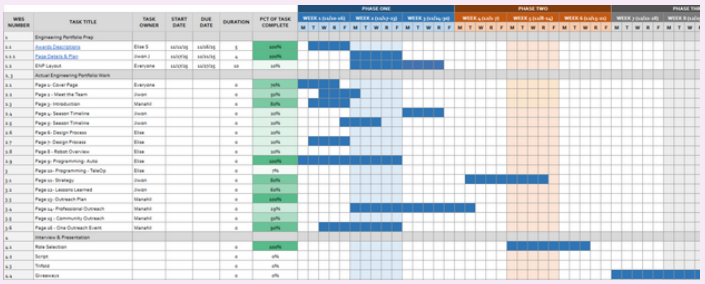
Key Statistics

- Outreach events completed: 14
- Competition performance: see page 3
- Member retention rate (x%) 120% increase (10 to 22)

Tracking System

We utilize tools such as shared documents and spreadsheets, more specifically, a Gantt chart. By using our Gantt chart, we can easily organize our tasks and due dates. Furthermore, we use the Kanbanchi software for day to day task management.

Gantt Chart



Schedule

In the first half of our season, we held weekly two hour general meetings, one hour subteam training sessions Mondays-Thursdays, Saturday meetings from 9am-3pm, and meetings over winter break. As we approached ILT, we transitioned to meetings on Mondays and Fridays from 4-7pm while maintaining our Saturday sessions.

SEASON TIMELINE Pg.5

At the onset of the **DECODE season**, we began brainstorming ideas to address the archaeological challenge. Through **teamwork and collaboration**, we **innovated, iterated and refined our robot** to prepare for each competition. After every event, we evaluated our **Pluses and Deltas**, allowing us to intentionally **improve our design, programming, teamwork, and work ethic** by learning from both **our successes and setbacks**. As the season progressed, we reached a turning point. After finishing second to last In Meet 1,, we recognized that our current approach and performance would not allow us to be competitive at ILT. So, **we adapted** by meeting outside of school, expanding into additional days and hours, committing ourselves to accomplish our goal. This shift improved our robot performance and helped us grow into a **more resilient team**.

TRIASSIC PERIOD	JURASSIC PERIOD	CRETACEOUS PERIOD	CENOZOIC ERA

MESOZOIC ERA

8/1 - 9/13 TRIASSIC PERIOD

- Team bonding activities to strengthen communication and collaboration
- Established team norms to promote accountability
- Analyzed the 2025-2026 Decode Season game
  - Developed **Wishes**, **Prefers**, and **Demands**
  - Reviewed the official game manual
  - Discussed match strategy and overall approach
- Brainstormed early-season robot design concepts
- Assembled an initial robot prototype (Barney v0)
- Began developing a CAD training curriculum with an **emphasis on DFMA**.
- Introduced programming to enhance robot functionality

9/14 - 10/20 JURASSIC PERIOD

- Conducted SoCal Showdown Outreach in collaboration with FTC 49, reaching **500+ Kids and parents!**
- Completed CAD training with a performance task
  - Highlighted **performance gaps** of Barney v0
    - Limitation of **ground intake** and **shooting accuarcy** plauged Barney V0
    - **Brainstorm** and Crayola CAD to test out different design archetypes
    - Design and test **individual subsystems** to ensure compatibility
    - Assembled Barney v1
- Introduce a new auto pathing software, **PedroPathing**, to enhance Autonomous Pathing efficiency

10/21 - 12/11 CRETACEOUS PERIOD

- Competed at Meet 1 with Barney v1
  - Highlighted issue between delay in handoff between CAD → Programming → Drive Team
    - Decide that Programming receives the robot for at least 1 week before comp
- Taste of Cams, Reaching out to **1200+** Students and Parents about FIRST and STEM!
- Consortium and Site Day outreach, reaching **900+ Students, Parents, and fellow FTC Teams!**
- Iterated upon **Barney v1** to remain competitive at **meet 2**
- Started brainstorming and Master Sketch of **Barney v2**
- Dodson Vex IQ Tournament outreach, reaching **250+ middle school students!**

12/12 - 1/25 CENOZOIC ERA

- Competed at Meet 2 with Modified Barney v1
- Finish CAD of Barney v2
  - Manufactured turret plate and side plates using CNC machines
  - Continued to make prototypes and iterate on Barney v2
- Establish the usage of Limelight 3A and Pinpoint for accurate autonomous and TeleOp shooting.
- Reached out to numerous professionals that gave valuable feedback on our robot and code.
- Cub Scouts and Magruder Middle School outreach, reaching out to **70+ elementary and middle school students!**
- Competed at Meet 3 with Barney Jr

PROFESSIONAL OUTREACH

Joseph Carpenter - 12/18/25



Who: Former Northrup Grumman engineer, extensive experience in design/ systems engineering.
How he helped: Mr. Carpenter reviewed our ILT robot CAD and gave us suggestions, like adding gussets for support and CADing our robot to make it more serviceable.

Froilan Sarmiento - 01/06/26



Who: MH-60R Powerplant Systems Maintenance Instructor.
How he helped: Being a mechanical engineer, Mr. Sarmiento helped us understand the concepts of gear ratios and how we would implement such ideas into our robot.

Dr. Jethro Virtusio - 01/12/26



Who: PhD in Computer Science, part-time Assistant Professor in department of Software Technologies at De Le Salle University.
How he helped: Dr. Virtusio gave us suggestions on our game strategy and reviewed our auto.

Julie Oh - 09/13/25



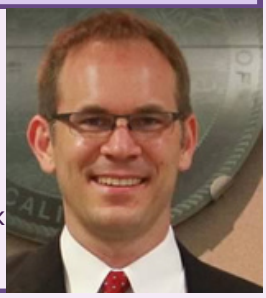
Who: Julie is the founder and CEO of Cutting Edge Innovation for 3 years.
How she helped: Ms. Oh taught the admin team how to utilize media tools like Instagram to spread our message. They used the skills they learned to create a reputation for our team online.

Travis Wu - 11/28/25



Who: FTC 21380 Bread Pandas alumni, currently attending UCLA, majoring in Data Science and Applied Math.
How he helped: Travis reviewed our portfolio and gave suggestions. We were able to use his advice and feedback to write our portfolio and present the essential information.

Eric Coe 12/15/25



Who: Eric has been an Aerospace Systems Engineer for RTX for 6 years.
How he helped: Mr. Coe reviewed our ILT robot design. He gave out considerations and solutions for potential problems for our ILT robot design. We used his feedback to iterate and improve our robot design.

Ryan Goy - 09/17/25



Who: Ryan is the founder/CEO of cadflow.ai.
How he helped: Mr. Goy taught the basics of 3D printing and critical thinking skills. We used these skills to create the 3D printed parts needed for our robot. It helped us a lot throughout the season as we were able to learn a new skill and fill our needs.

Edison Sarmiento - 01/08/26



Who: MM1(SW) Sarmiento, Assistant Leading Petty Officer, Engineering Shop, ACU 1.
How he helped: He gave us general feedback on our robot CAD and shooter CAD. He advised us to lighten our shooter design to improve its accuracy. We implemented his advice in our robot for meet 3 and succeeded.

Recruiting Mentors

Mr. Lim



How he helped: As an experienced mentor, he provided invaluable insight and guidance to improve our robot and team structure.

Mr. Umamaheswaran



How he helped: Having numerous years of experience as a software engineer, he provided valuable insight on code structure and organization, along with version control.

Mr. Hildebrandt

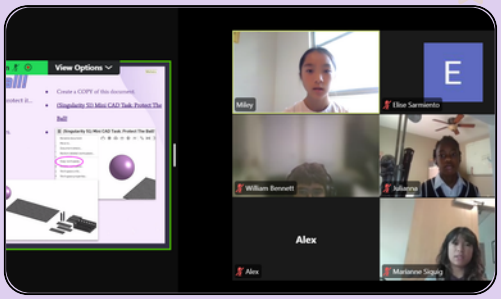


How he helped: His technical expertise in Product Development provided guidance on how to systematically improve the robot design while meeting a demanding schedule.

COMMUNITY OUTREACH

CAD Workshop with Team 25370 - 09/27/25

Because many of our **current** FTC members previously competed in **Dodson Middle School FTC Team, RoboDolphins 25370**, we hosted a **CAD workshop**. During the workshop, our members taught members of RoboDophins some fundamental CAD skills, including **constraints** and **dimensioning**, the use of **assemblies**, and **subsystems**. Students were also assigned hands on design challenge to create a **design** that can fully secure the ball from all sides using **CAD**. To support continued learning, we shared instructions on how to access the **FTC parts Library** and additional **CAD resources**. Ball activity Design Challenge : After the workshop, our CAD team challenged students of RoboDophins25370 to create a design that can fully secure the ball from all sides using ONLY the parts provided and FASTENED mates.



SoCal Showdown - 10/11/25

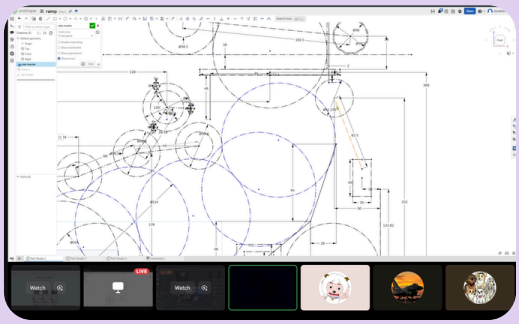
At **SoCal Showdown**, we reached over 250 **elementary** and **middle school** students through targeted **STEM outreach.** We facilitated four hands-on activities which included **cookie carving** to introduce precision and design concepts, **spaghetti towers** to promote structural thinking and teamwork, and **robot driving** to simulate FTC competition skills from our **team drivers**. These activities increased participants' **confidence**, **communication skills**, and **interest** in **FIRST programs**, helping to **inspire sustained engagement** STEM fields. Cookie Carving: Students would carve chocolate chips out of cookies then eat it after. Spaghetti tower: We **challenged** students to create a 2 feet tall tower out of spaghetti and marshmallow in two minutes. Origami challenge: We would **teach** students how to make origami then If students could remake the origami, they get candy.



People reached: 300+
Target audience: 2nd - 8th graders

SoCal Alliance outreach - 10-02/25-26

Team 51 **collaborated** with Teams 30099 (OmicronX), 15046 (Incognito Robotics), and 27353 (Blue Lancers) under the SoCal Alliance, meeting **weekly** to share **advice on management, design, and strategy** while **tracking a total of 73 collaborative hours** this season. Focused primarily on **design** and **CAD mentoring**, Team 51 supported 27353 in **developing** their **turret** and **shooter systems,** helping them integrate an X-contact bearing, **refine their design criteria**, and providing CAD **guidance** for a **more efficient layout.** With Team 51's assistance, 27353 completed their robot CAD by January 3rd and **fully assembled** it by January 20th, giving their programming team two weeks to prepare for their ILT competition.



CAMS Site Day - 11/15/25

At **CAMS Site Day**, an annual event where future freshmen and families explore the campus, our robotics team showcased **competitive FTC robotics** to over 400 middle school students. Visitors observed the **design process** firsthand and learned how **programming**, **engineering**, and **teamwork** come together in competition. Team members also engaged prospective students in conversation about **joining** the FTC program in high school, resulting in over **30 committed recruits** for next season. Handed out brochures: We handed out flyers that contained info about our team's mission and vision statement. Scrimmage invitations: Our team invited visiting students to our **future FTC meets**. Robot Demo: Our drive team presented out current robot and showed students' our progress so far.



Target audience: 8th grade
People reached: 500+

COMMUNITY OUTREACH

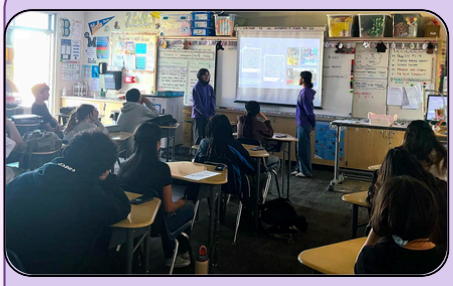
Dodson VEX IQ Tournament - 11/19/25



Target audience: 5th - 8th grade
People reached: 250+

At the **VEX tournament**, the FTC 51 booth engaged over **250 middle school students** passionate about robotics, presenting **FIRST Tech Challenge** as a natural progression from **VEX IQ**. We **emphasized** the robot design, programming, and competition of **advanced robots** as an opporunity to apply teamwork, creativity, and **engineering skills**. Many CAMS high school applicants from Dodson, expressed strong interest and **enthusiasm** in continuing their robotics journey with FTC., resulting in significant **recruiting outcomes**. This initiative effectively transformed student's **enthuiasm** into **prospective team members,** advancing our program's growth and sustainability. Robot demo: Our team presented our robot to not only Dodson's Vex team but also their own FTC team. We showed them how far along we were as well as what they would be doing should they graduate and join team 51.

Magruder Middle School Outreach - 01/16/26



Target audience: 6th- 8th grade
People reached: 40+

Our team visited Magruder Middle School to introduce more than **40 students** to the exciting world of the **FIRST Tech Challenge** . We shared what FTC is, our **team's story**, and how middle school students begin their own **robotics journey**. Our team members highlighted the many benefits of joining FTC, including learning programming and **developing** engineering skills, strengthening **teamwork, problem-solving, and leadership abilities**. We also presented this season's game, **DECODE**, themed around archaeology and artifact discovery. Students **learned** how teams score purple and green artifacts into red and blue alliance goals, and how each year's **unique game challenges** teams to design, build, and program robots to complete complex tasks under time constraints. By the end of the visit, many students were **excited** about joining **future** FTC teams and exploring how robotics can bring **STEM** to life.

Cub Scouts Pack 413 Outreach - 01/12/26



Target audience: K - 6th grade
People reached: 30+

Our team hosted an engaging outreach event with **Cub Scouts Pack 413.** We introduced **STEM** and **robotics through FTC.** We began with a presentation to the scouts explaining **what FTC is and how it inspires students worldwide** to design, build, and program their very own robots, while also sharing **our team's progress, goals, and experiences**. We then brought these concepts **to life with hands-on robot demonstrations** to see FTC robots in action. In the end, the scouts gained early exposure to **STEM** concepts **through robotics** and **inspired them to join FIRST Programs** in the near future. Our team presented our robot to the scouts and allowed them to get hands-on experience with it. Scrimmage Invitation: We invited the scouts and their parents to come support us in our future competitions.

LA Makers Faire - 04/26/2026

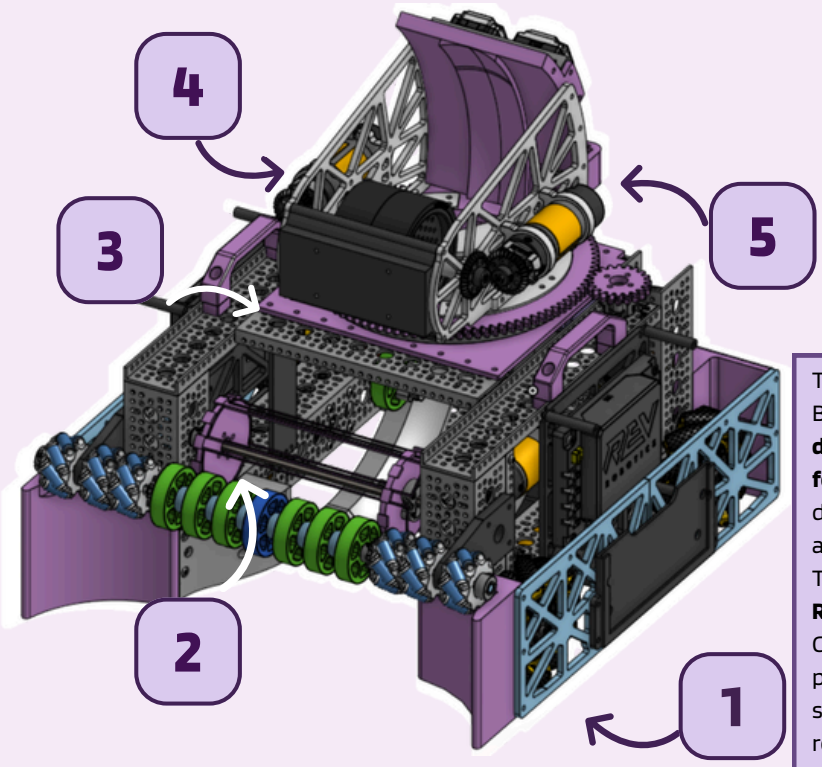


Target audience: Our Local STEM Community
People reached: 500+

At **LA Makers Faire** our team hosted a booth dedicated to sharing more about the **FIRST** program to our **local STEM community**. Our team highlighted how FTC creates a space where students can **develop** their **teamwork, leadership, and STEM skills** in preparation for professional work environments. There we taught students how to drive and control a sphere robot, presented to parents of students **what FTC is** and **how to get involved** with their **own local robotics team**. We also held a fundraiser for our **Canada cup premier event** by selling team merch such as key chains, 3D printed Dinosaurs, sticker, and pins. After introducing **FIRST as a program**, we then showcased what our team has done with the season through holding hands on **robot demonstrations**, displaying our **engineering portfolio**, as presenting our 226 Point Solo Run.

ROBOT OVERVIEW Pg.9

PRESENTING: BARNEY JR.



SUBSYSTEM OVERVIEW

1. Modified goBILDA Drivetrain
2. Rubber Band & Compliant Wheel Intake
3. Gear-Driven Lazy Susan Turret
4. Fixed Angle Hood Shooter
5. Gate opener

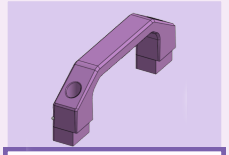
OVERALL ROBOT DESIGN

Throughout the season, we "**excavated**" and refined multiple subsystems on Barney Jr., continuously improving overall performance. We began with a **drivetrain** derived from the goBILDA Starter Kit, which **provided** a **reliable foundation** during early-season meets. As our **understanding** grew, we designed and integrated a **rubber-band intake** that **efficiently** collects artifacts, **improving consistency** in later matches.

To enhance scoring, we developed a **gear-driven turret** powered by a **1,150 RPM** motor with a **5:1** gear ratio, enabling **precise** and **controlled aiming**. Complementing this system is a **gear-driven, fixed-angle hooded** shooter powered by **two 6,000 RPM** motors. This design enables **accurate**, **rapid-fire** scoring and **significantly** reduces cycle time. Together, these subsystems reflect our **iterative** design process and our focus on **mechanical efficiency** to achieve **competitive** success.

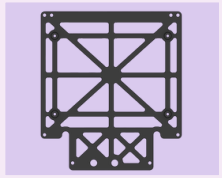
DESIGN PHILOSPHY: DESIGN FOR MANUFACTURING AND ASSEMBLY

MISCELLANEOUS:



ROBOT HANDLES

Applying DFMA principles we recognized that the robot's size and weight made transportation difficult. We designed a mounted ergonomic carrying handles to ease transporting the robot.

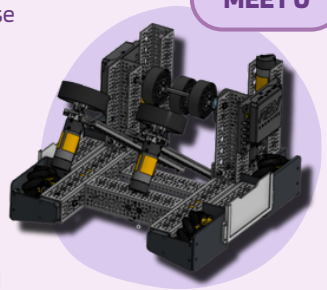


HUB MOUNT

We designed a **custom hub mount.** This enabled us to be flexible in mounting **control systems on the robot's** drivetrain, providing a custom fit, and assisting with clean and safe wire management..

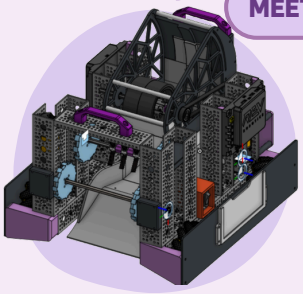
BARNEY JR.'S EVOLUTION

MEET 0



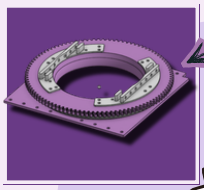
Horizontal flywheel shooter was inconsistent and scored minimal points.

MEET 1-2



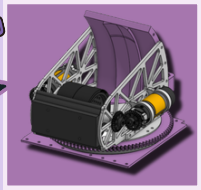
Fixed angle hooded shooter with kicker transfer. Kicker mechanism was inconsistent.

SHOOTER/TURRET:



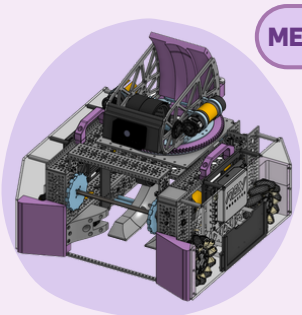
Mounting holes for attaching the shooter.

Detachable Shooter



With DFMA in consideration, we made significant improvements to both **shooter and turret systems.** Our turret is designed to be **modular,** allowing it and the shooter to be removed as a single assembly. Turret includes integrated gusset mounts for the shooter to be easily bolted on, **simplifying assembly and maintenance.** This modular design makes repair more efficient and enables faster prototyping to test new shooter or turret configurations.

MEET 3-ILT

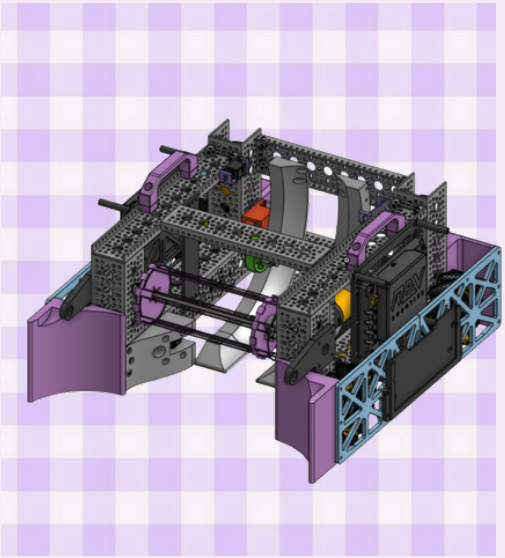


Turret shooter for faster firing sequence.

DRIVEBASE Pg.10

DRIVEBASE OVERVIEW

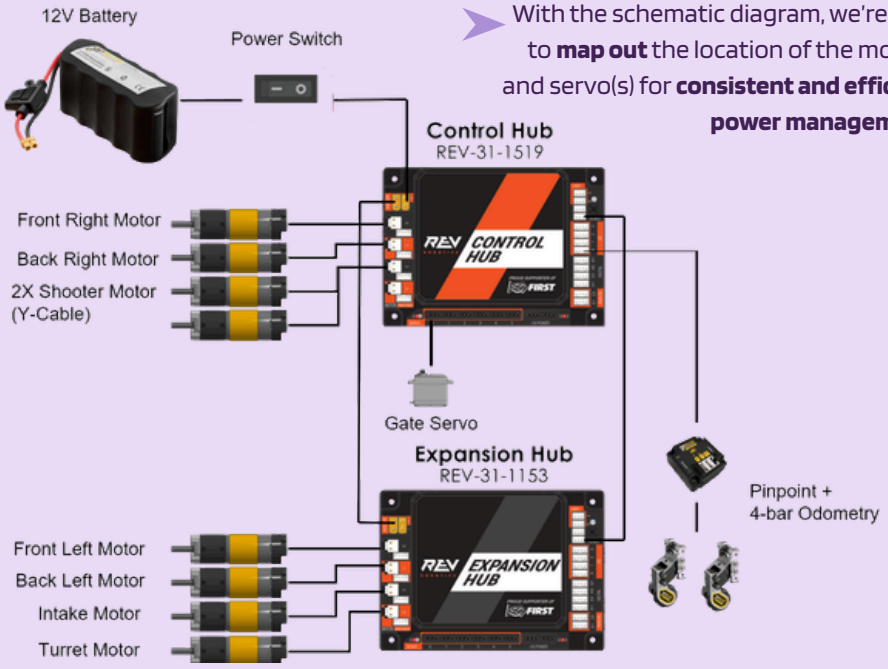
CHASIS



KEY FEATURES

- **4x 435 RPM Motors Drive base to allow the robot to travel quicker and better maneuverability to reach artifacts faster**
- **16.4in x 16.4in for endgame parking**
- **Miter gear drivebase setup 1:1 ratio to allow smoother drive**
- **Modular Mount of Electronics to allow easier access for maintanence**

SCHEMATIC DIAGRAM



- With the schematic diagram, we're able to **map out** the location of the motors and servo(s) for **consistent and efficient power management**.

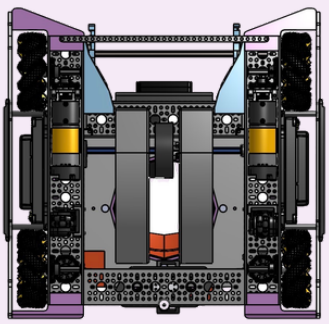
Wheels

Problems

- Smaller Miter gear prone to skipping/jamming
- Motor placement was inefficient

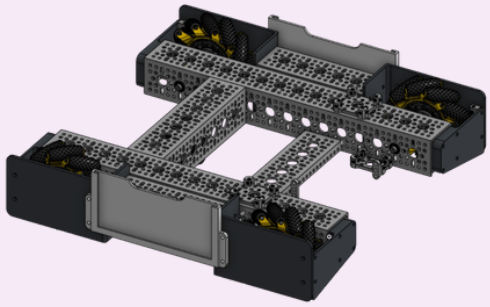
Solutions

- Used wide miter gear to resolve jamming
- Placed motor inside U-channel without compromising access



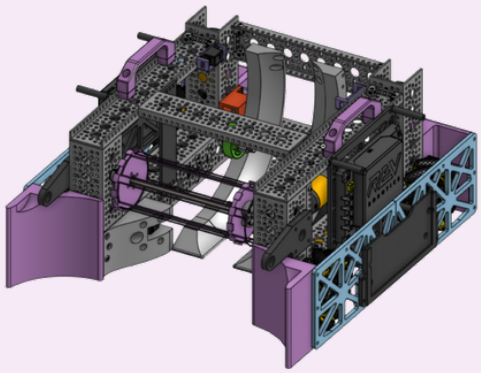
Drivebase Evolution

Barney V0-V2 Chasis



Problems

- The chassis required continuous **maintenance** to avoid **performance loss**
- Previously installed smaller miter gear, but it would sometimes skip, making it inefficient .
- The wheel guard is **prone to breaking** during competition when **facing heavy defense**



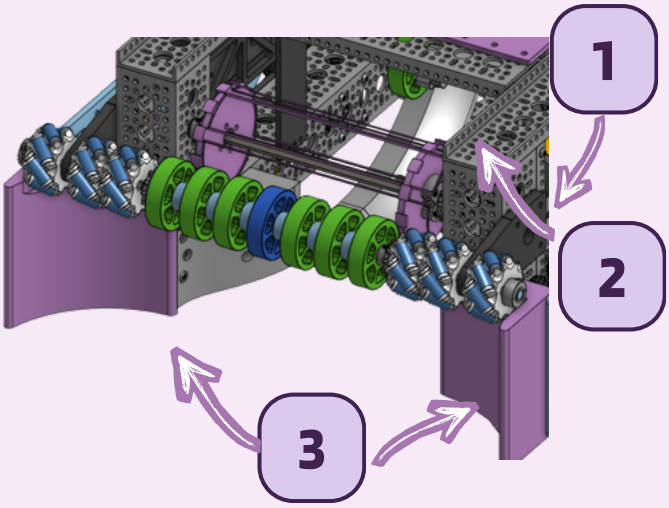
Barney Jr Drivebase

Implemented Solutions

- **3D printed** drivetrain outer plate provide **protection while being lightweight.**
- Changed bevel gears from small to its **wider version,** making it more driver-friendly.
- Made wedges to **protect** wheels and to improve **intake**

INTAKE/INDEXER Pg.11

Intake Overview



Key Features

- Rubber-band based intake
- Metal bar for the support of storage
- Ramp for ball **storage** + **direction**.
- Supplemental Gecko wheels indexer to **eliminate dead zones, flexible compression**, and **rapid shooting**.
- Curved Wedges for **funneling**

1

Intake extension spanning **15 inches,** allowing increased funneling compared to the previous iteration. Matches with **"Touch it**, **Own it Strategy"** allowing consistent contact and full control of the artifact.

2

A single-wheel intake with custom ramp **eliminated** dead zones, while supplemental Gecko wheel improved artifact collection and **shots-per-cycle efficiency** by **30%**

3

Twin curved wedges are attach to our drivetrain outer plate. These wedges help funnel artifact into our rubber band intake. This prevents jamming, reduces our cycle time from **10 second → 7.3 seconds** and Increasing intake speed by **23%**

Engineering Process in action: Integrated Indexer

Indexer: Resolves Dead Zone - This system solved our issue where intake did not move the 3rd artifact to the shooter.

Solution Iterations

Version 1
Kicker was designed, powered by a servo, to move artifacts **directly into the shooter**.

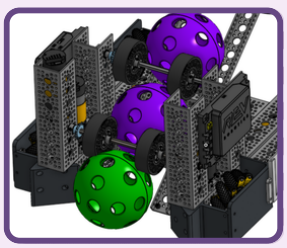
Version 2
Initial kicker lacked speed. We switched to a **Gecko wheel indexer**, which **significantly improved speed and enabled rapid shooting**

Version 3
Initial indexer was inconsistent. **Surgical tubing** was added on **each side** of the intake to improve **consistency**.

Version 4
Initial indexer is now consistent. **Surgical tubing, vector,** and **complient** wheels was added on to imporve the **range** of our intake.

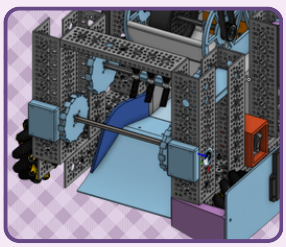
REITERATIONS

V1
2x 72mm Gecko Wheels



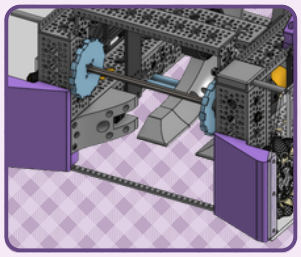
PROS	CONS
• Early iteration of our active intake. • Simple intake utilizing strictly goBILDA parts.	• No proper ramp for guidance. • Deadzone between the second stage of intake & shooter.

V2
2 rubber-band wheels



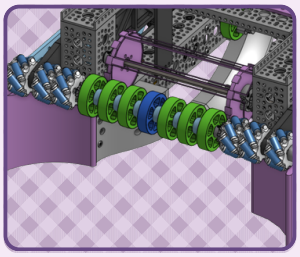
PROS	CONS
• Curved ramp to accommodate ball guidance. • Consistent intake.	• Geometry did not allow rapid firing. • Prongs on intake hoists would often break.

V3
1 Rubber-band wheels, custom ramp



PROS	CONS
• Intake + ramp allows rapid firing. • Reduced complexity.	• Intake isn't wide enough (Periodically jams when 2 balls are in.)

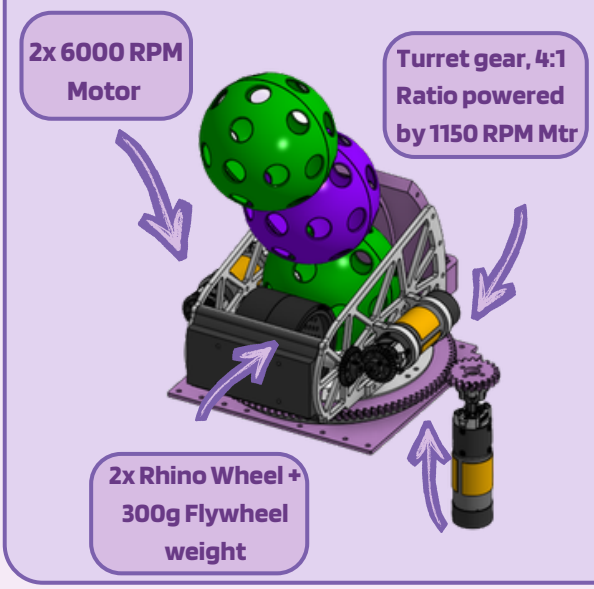
V4
1 Rubber-band intake & extension w/ vector wheels



PROS	CONS
• Intake + ramp allows rapid firing. • Being able to collect balls smoother and quicker.	• Easier to break • Harder to fix and replace

SHOOTER/TURRET Pg.12

SHOOTER AND TURRET OVERVIEW

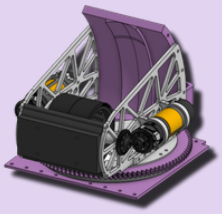


Key Features

- **90%** printed parts (PLA filament) to **fail fast**, **succeed faster** when **prototyping**
- Turret allow **shooting** from **any angle**
- **2X 6000 RPM motors** to maintain **effiency** and **consistent velocity** when **rapid firing**
- 550g Flywheel to maintain interia

UTILIZATION OF SHOOTER & TURRET

We observed that most **cycle time was spent aligning the shooter** during competition. Prior design also lost **velocity when rapidly firing artifacts**.
Adding a **turret** & **second motor** enables:

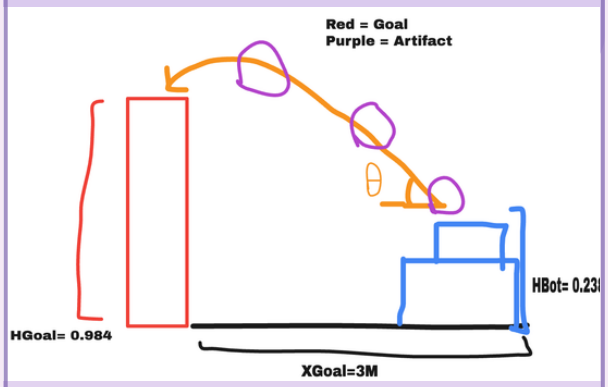


- Score **any position** in **any launch zone**
- Rapidly fire **3 artifact in 0.23 seconds**
- Decrease cycle times by **2 seconds**

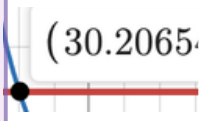
Before Turret: 27 Seconds for **9** Artifact, **70%** relability
After Turret: 25 seconds for **12** Artifact, **95%** Realbility

Design Goal: **Rapidly shoot three artifact** without premature firing while achieving **>= 90% accuracy from close range** and **>= 85% accuracy** from Far Launch Zone.

SHOOTER ANGLE MODEL

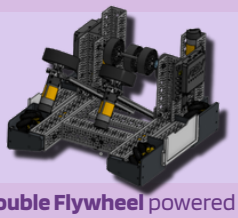


Before **designing** our Shooter, we outlined a **key requirement**: the ability to shoot from the **Far Launch Zone at a fixed angle.** We first created a simple diagram to represent our goal. Then, using **Projectile Kinematic Equations**, we modeled a **3 meters (XGoal)** shooting distance with a goal height of **0.984 meters (HGoal),** accounting for robot **height of 0.238 Meters (HBot).** By expressing **Ymax** as the **height of the ball need to rise,** which calculated by **subtracting** the **height of the goal from the height of the robot. Xmax** represents the **minimum distance from the goal.** We simulated the solution in Desmos, resulting in optimal launch angle of **30.2 degree** for consistent scoring from **3 Meters!**



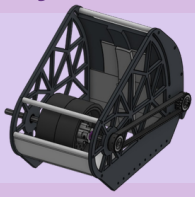
$$t_{max} = \frac{v_0 \sin(\theta)}{g}$$
$$x_{max} = v_0 \cos(\theta) t = \frac{v_0^2 \sin 2\theta}{2g}$$
$$v_0^2 = \frac{2 y_{max} g}{\sin^2 \theta}$$
$$x_{max} = \frac{2 y_{max} g \sin 2\theta}{2g \sin^2 \theta} = \frac{y_{max} \sin 2\theta}{\sin^2 \theta}$$

Barney 1 V1 Shooter



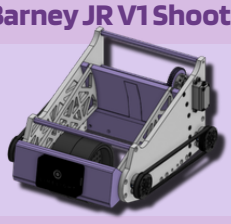
- **Double Flywheel** powered by **2x 6000 RPM** motor angled at **22.5 Degree**
- **Easy to build**, using **COTS** parts
- **Inaccurate** when shooting
- Lost **velocity quickly** when fired **rapidly**

Barney 2 V1 Shooter



- 2x 96mm flywheel powered by 1x 6000 RPM motor
- **Accurate** when **shooting** from **far** and **close launch zone**
- **with a fixed X angle**
- Lost **velocity** quickly if fired **rapidly**

Barney JR V1 Shooter



- **2x 96mm** flywheel (**300g**) powered by **2x 6000 RPM motor**
- **Accurate** when **shooting** from **far** and **close launch zones**
- **Adjustable angle** via **rack** and **pinion**, driven by **servo**
- **Maintained** Velocity if **rapidly fired**

Barney JR V2 Shooter



- 2x 96mm flywheel + 300g weight powered by 2x 6000 RPM motor
- **Accurate** when **shooting** from **far** and **close launch zones**
- **Simple Fixed X angle Versus Complex Adjustable**
- **Maintained Velocity if rapidly fired**

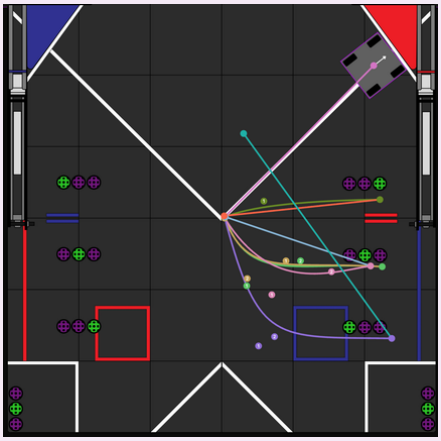
PROGRAMMING

SOFTWARE OVERVIEW

Our code is organized into **modular subsystems** (Shooter, Turret, Intake, Drivetrain) that operate independently. State machines manage **autonomous** sequences with **sensor-based** transitions and timeout failsafes, replacing unreliable timing-based approaches. This architecture allows us to tune **individual components** without affecting other systems.

AUTONOMOUS

For autonomous navigation, we use **Pedro Pathing** - a reactive path follower that **continuously corrects** for **positional** and **rotational error** using three PIDF controllers. Unlike pre-programmed motion profiles, Pedro dynamically adjusts based on real-time odometry feedback, recovering from collisions without human intervention.



Auto Path for our 18 artifact autonomous routine, which has two gate cycles

CUSTOM VELOCITY CONTROL

The Challenge: REV's built-in PIDF uses a fixed F constant that does not scale with motor velocity. At different shooter speeds, the controller behaved inconsistently, with a slow spin-up and poor recovery after each shot.

We replaced it with a three-term custom controller using direct power output:

$u = kS + kV \cdot v_{target} + P \cdot (v_{target} - v_{current})$

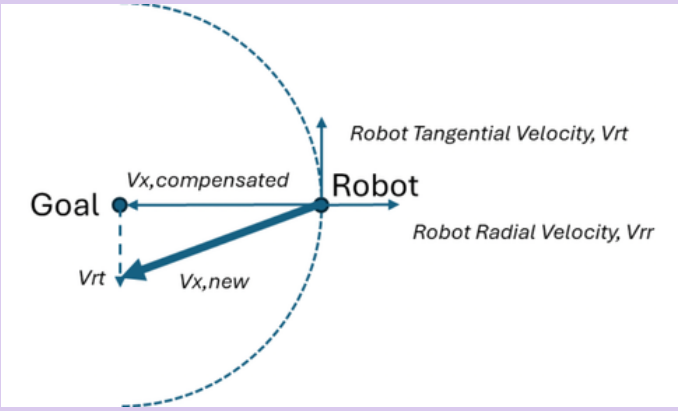
kS overcomes motor stiction. kV · v_target scales feedforward power linearly with desired speed, and P · error corrects for disturbances. Now, we have a spinup time of **0.2s** (from 3.5s), and extremely accurate rapid fire

SHOOTING ON THE MOVE

The Challenge: At full drive speed, shots consistently miss. The ball's trajectory doesn't account for the robot's motion, causing under/overshoot depending on direction.

Our solution: Using Pinpoint V2 odometry, we read the robot's velocity vector each loop and decompose it into two components relative to the goal:

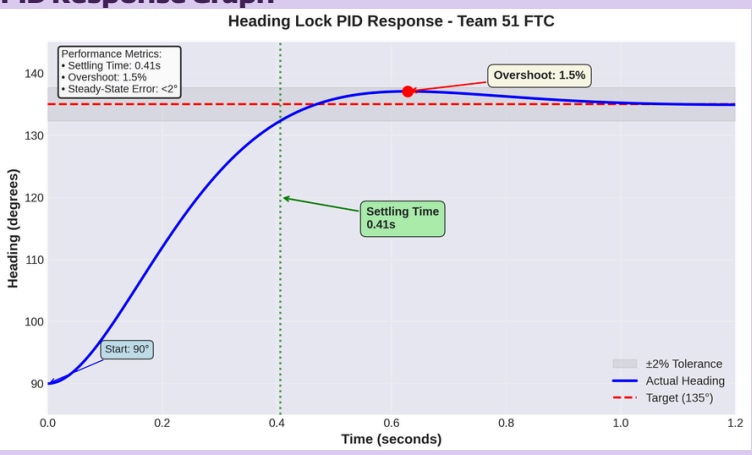
- Radial (toward/away) → adjusts flywheel RPM
- Tangential (sideways) → offsets turret lead angle



HEADING LOCK

On our V2 Design with no turret, we implemented a heading lock system, which uses a PID controller to automatically snap to an angle facing your alliance goal.

PID Response Graph



Result: 23% faster scoring (87->101 points in solo teleop)

PROGRAMMING

TURRET-AUTOTRACKING

Manual aiming wastes time, so we built automatic turret rotation. Instead of using a Limelight, we opted to using Pinpoint v2 odometry as odometry proves to be faster and more reliable, and vision can get blocked. Pinpoint gives continuous 1500 Hz updates regardless. We upgraded from 3-wheel odometry for this, gaining more precise and accurate localization and built-in IMU fusion.

The Process

Using odometry, we calculate two values: the angle to aim the turret and the distance for our shooting lookup table.

$$\theta = tan^{-1}\left(\frac{y_{goal} - y_{robot}}{x_{goal} - x_{robot}}\right)$$

This gives us the angle from robot to basket in the field frame. We convert to robot frame by adding our heading, then apply a offset for our turret's zero position. A PID controller rotates the turret to this angle with a 2-tick deadband to prevent oscillation.

$$distance = \sqrt{(y_{robot} - y_{goal})^2 + (x_{robot} - x_{goal})^2}$$

This distance feeds directly into our shooting lookup table, which returns the correct RPM for that range.

Below is the code for the aimAtGoal method

```
public void aimAtGoal(double robotX, double robotY, double robotHeading, boolean isBlue) {
    // Rotate turret offset by robot heading
    double cos = Math.cos(robotHeading);
    double sin = Math.sin(robotHeading);

    double rotatedOffsetX = TURRET_OFFSET_X * cos - TURRET_OFFSET_Y * sin;
    double rotatedOffsetY = TURRET_OFFSET_X * sin + TURRET_OFFSET_Y * cos;

    // Calculate turret position in field coordinates
    double turretX = robotX + rotatedOffsetX;
    double turretY = robotY + rotatedOffsetY;

    // Goal basket positions (field coordinates)
    double goalX = isBlue ? 0 : 144;
    double goalY = 144;

    // Calculate angle from turret to goal (field-relative)
    double angleToGoalRadians = Math.atan2(goalX - turretX, goalY - turretY);
    double angleToGoalDegrees = Math.toDegrees(angleToGoalRadians);

    // Convert robot heading to degrees
    double robotHeadingDegrees = Math.toDegrees(robotHeading);

    // Calculate turret angle relative to robot
    double turretAngle = angleToGoalDegrees + robotHeadingDegrees - 90;

    // Normalize to -180 to +180
    turretAngle = normalizeAngle(turretAngle);

    // Set the target angle (will be clamped to limits)
    setTargetAngle(turretAngle);
```

DISTANCE-BASED SHOOTING

The Challenge: Shooting from a fixed angle and velocity only works at one optimal distance. To score consistently from anywhere on the field (1.0m - 3.0m from basket), we need to dynamically adjust **flywheel velocity based off the robots position on the field**

Our Initial approach used physics formulas - range = $v_0^2 \sin(2\theta)/g$ - however this didn't account for ball compression, hood geometry, and the ball's spin

The **new** testing method we used is:

1. Positioned the robot at **measured distances** (ex: 1.0m, 1.5m, 2.0m from basket)
2. **Test multiple** RPM combinations at each distance
3. **Record** all pairs that consistently make the basket (>80% over 10 shots)

Trial #	Distance (inches)	Velocity (RPM)	Predicted RPM	Error (RPM)
1	73.2	1160.0	1070.6	89.4
2	92.8	1090.0	1167.7	-77.7
3	101.9	1240.0	1212.7	27.3
4	107.8	1110.0	1241.9	-131.9
5	119.0	1290.0	1297.4	-7.3
6	129.3	1350.0	1348.3	1.7
7	142.5	1370.0	1413.7	-43.7
8	153.4	1480.0	1467.6	12.4
9	153.9	1600.0	1470.1	129.9

Using these points, we used a 2D lookup table and a linear interpolation formula to account for every position on the field.

LIMELIGHT

The Challenge: Pinpoint v2 odometry drifts over time due to wheel slip and collisions, causing the turret to gradually miss the basket by increasing margins (5-10° by endgame).

Our solution: The Limelight 3A with MegaTag2 AprilTag detection provides constant relocalization, resetting accumulated drift without disrupting odometry tracking.

Our Implementation:

Static configuration: Fixed limelight position provides consistent AprilTag visibility regardless of turret orientation, eliminating turret angle tracking complexity

Confidence-based corrections: Validates detections before applying - requires high quality (decisionMargin > 35) and rejects jumps larger than 24 inches

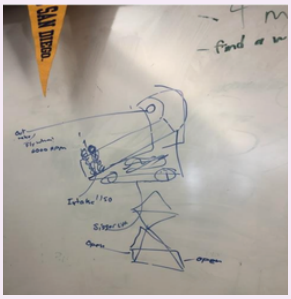
Robot orientation tracking: Updates Limelight with current heading each loop to transform camera-relative detections into field coordinates

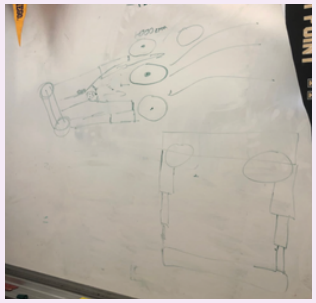
DESIGN PROCESS

DESIGN STRATEGY

1) Evaluate

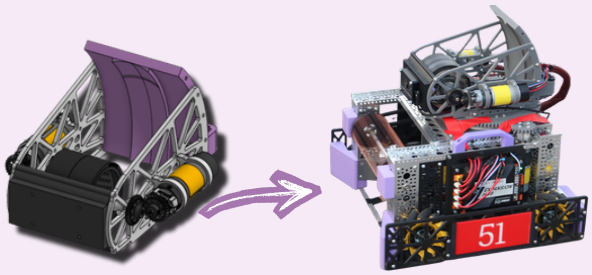
As a team, we reviewed the **game manual** and **reveal video** to explore **potential** scoring opportunities and assess our team's **priorities**, **strategies**, and **core objectives.** This process helps us **formulate** an effective match approach while setting clear goals for **performance** and **improvement**.





3) Design & CAD

Our goal for this season is to approach **design** & **CAD with DFMA** (Design for Manufacturing and Assembly) **Mindset**. To do this we initially Crayola CAD basic geometry to test different **archetypes** and their **feasibility**. This allow us to transfer the design into a Master sketch, and create parts for the robot. While designing the parts we make sure to include **modularity** and have **hardware** be easily **accessible**!



5) Analyze & Reiterate

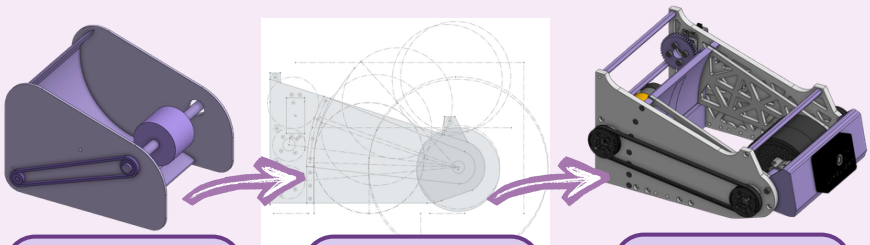
Based on the data we collected, we **persevered** by coming up with various ways to improve our designs in the end until it can finally **accommodate** our robot's needs, whether it'd be **modifying the CAD or utilizing version controls.**

Wish, Prefer, Demand System

Wish	Prefer	Demand
Lift mechanism for Endgame	Turret to allowed 360 Degree Scoring	Ability to hold 3 artifacts
Spindexer/Linear Indexing	Wide Intake	Intake anywhere on field
Triple Shooter	Custom drivetrain	Ability to shoot and score at Far and Close Zone

2) Brainstorm

We start brainstorming session by breaking into **pairs** to create a design that fit our **goals** and **criteria's**. Then each pair share out their ideas to the team by highlight the **capabilities** and potential **flaw**. Brainstorming allow us to **rapidly** share ideas that can be **transfer** into **Crayola CAD**

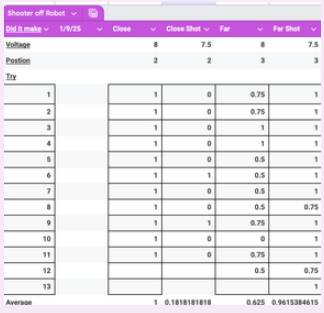


Crayola CAD | Mastersketch | CAD

4) Prototype & Implement

Testing our **designs**, trial and error played a **significant** role in our **iterative** process. We often made rough assemblies, to which we would then run to collect data. We asked ourselves: "**What was supposed to happen? What actually happened?**" and **"How we can we improve from the data we collected?**"

DATA DRIVEN ITERATIONS



Shooter off Robot					
Did it make	1/9/25	Close	Close Shot	Far	Far Shot
Voltage		8	7.5	8	7.5
Postion		2	2	3	3
Try					
1		1	0	0.75	1
2		1	0	0.75	1
3		1	0	1	1
4		1	0	1	1
5		1	0	0.5	1
6		1	1	0.5	1
7		1	0	0.5	1
8		1	0	0.5	0.75
9		1	1	0.75	1
10		1	0	0	1
11		1	0	0.75	1
12				0.5	0.75
13					1
Average		1	0.181818182	0.625	0.961538462

Motor testing revealed that our **design** utilizing GT2 timing belt caused **significant efficiency loss** during shooting, requiring a **redesign** using **miter gears f**or power transfer.

